
Example: Logistic Regression. AMS 580

Name: _____ SBU ID: _____

Dear all, this example will teach you the necessary R procedures for logistic regression that you will need for the upcoming homework and quiz.

Logistic Regression (*A type of **Generalized Linear Model** when the response is binary) with the **bwt Data**

The PimaIndiansDiabetes2 [in `mlbench` package] data is a built in R dataset containing 9 variables and 768 cases - please see the following link for detailed description:
<https://rdrr.io/github/quantide/qdata/man/pimaindiandabetes2.html>

Your task is to use all the other 8 variables to predict the binary dependent variable 'diabetes' telling us whether the subject is diabetic or not (factor with 2 levels: `neg` and `pos`). You will split the data into 80% training and 20% testing, using `seed = 123`, and follow the procedures from the following website:
<http://www.sthda.com/english/articles/36-classification-methods-essentials/151-logistic-regression-essentials-in-r/>

- (a) Please split the data into 80% training and 20% testing using `seed = 123`.
- (b) Then you shall fit a logistic regression model with all the other 8 predictors using the training data.
- (c) Please use this fitted model based on the training data to predict the response variable 'diabetes' (whether the subject is diabetic or not) for the testing data. Please generate the confusion matrix, and report:
 - (i) The overall accuracy;
 - (ii) The sensitivity (that is, the probability a subject is predicted to be diabetic given that he/she was in fact diabetic);
 - (iii) The specificity (that is, the probability a subject is predicted to be not diabetic given that he/she was in fact not diabetic).

